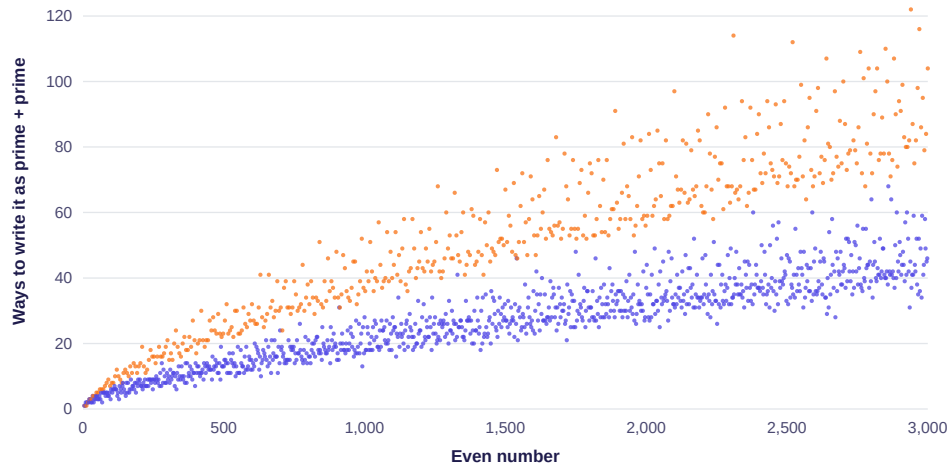


Goldbach's Conjecture

Every even number bigger than 2 can be made by adding two primes.



Each dot is an even number and how many ways it splits into two primes. The count keeps growing, and never hits zero.

TRY IT

$$4 = 2 + 2$$

$$10 = 3 + 7 = 5 + 5$$

$$28 = 5 + 23 = 11 + 17$$

$$100 = 3 + 97 = 11 + 89 = 17 + 83 = 29 + 71 = 41 + 59 = 47 + 53$$

WHAT MATHEMATICIANS KNOW

- Christian Goldbach proposed it in a 1742 letter to Leonhard Euler.
- Computers have checked every even number up to 4 billion billion (4×10^{18}). No exceptions.
- In 2013 Harald Helfgott proved a “weak” version: every odd number greater than 5 is the sum of three primes.

WHY IT'S STILL UNSOLVED

Checking numbers, no matter how many, is evidence, not proof. The picture suggests bigger numbers have more and more ways to work, which makes a failure seem very unlikely. But no one has found an argument that covers every even number at once.

TRY THIS WITH STUDENTS

- Give each group a different even number to split every possible way. Pool the results into a class table and graph it: you have built your own “comet.”
- Ask: Which even numbers have the most ways? (Look at multiples of 6.)
- Ask: Why only even numbers? What goes wrong with odd ones?

WORDS TO KNOW

Prime a whole number greater than 1 whose only factors are 1 and itself

Conjecture a statement believed true but not yet proved

Online, with the other seven problems: ohiomathteacher.github.io/applet-library/open-problems/goldbach.html · The picture was computed from the real numbers.